TDD Test Report

Smart Cold Box — Sprint 2 | Story 004-1: Accumulate Heat Exposure (Session 2: `accumulate_heat()`)

> **Suite Summary:** Executed 6 test cases for state accumulation. All 27 tests in the full test suite (`heat_exposure/test_heat_exposure.py`) passed cleanly (*Ran 27 tests in 0.0025s — OK*).

Test Case Summary Table

Feature	Expected Output	TC #	Test Case Name	Dev	Date
004-1 Accumulate heat exposure	`{"heat_s": 0, "last_good": None}`	**TC-07**	New state starts at zero	Kristranan P.	10 Nov
	`heat_s` stays 0; `last_good` = first reading	**TC-08**	First reading remembered & adds nothing	Kristranan P.	10 Nov
	`heat_s` goes 600 → 900 s (9.0°C at 300s, next at 600s)	**TC-09**	Total grows by heat time	Kristranan P.	10 Nov
	`heat_s` stays 0; `last_good` stays 9.0°C reading	**TC-10**	Failed read keeps last good reading	Kristranan P.	10 Nov
	Same as new state: `heat_s` 0, `last_good` = reading	**TC-11**	Missing state starts a new total	Kristranan P.	10 Nov
	Old state still `heat_s` 0; new state `heat_s` 300	**TC-12**	Old state is not changed	Kristranan P.	10 Nov

TC-07: New state starts at zero (`test_new_state_starts_at_zero`)

Expected Output: `{"heat_s": 0, "last_good": None}`

RED PHASE

Wrote test first. `new_heat_state()` did not exist, stopping with `AttributeError`.

```
def test_new_state_starts_at_zero(self):
    self.assertEqual(hx.new_heat_state(), {"heat_s": 0, "last_good": None})
```

```
$ python -m unittest -v test_heat_exposure
test_new_state_starts_at_zero ... ERROR
AttributeError: module 'heat_exposure' has no attribute 'new_heat_state'
FAILED (errors=1)
```

GREEN PHASE

Smallest code that passes: return dict with `heat_s = 0` and no reading.

```
def new_heat_state():
    return {"heat_s": 0, "last_good": None}
```

```
test_new_state_starts_at_zero ... ok - OK
```

YELLOW PHASE (REFACTOR)

Added docstring defining start state semantics. Tests still pass.

```
def new_heat_state():
    """Start of a shipment: no heat time yet, no reading yet."""
    return {"heat_s": 0, "last_good": None}
```

TC-08: First reading remembered and adds nothing (`test_first_reading_is_remembered_and_adds_nothing`)

Expected Output: `heat_s` stays 0; `last_good` = initial reading

RED PHASE

`accumulate_heat()` did not exist yet.

```
def test_first_reading_is_remembered_and_adds_nothing(self):
    r0 = reading(0, 9.0)
    state = hx.accumulate_heat(hx.new_heat_state(), r0, LIMIT)
    self.assertEqual(state, {"heat_s": 0, "last_good": r0})
```

GREEN PHASE

Store record into `last_good` and return updated state dictionary.

```
def accumulate_heat(state, record, upper_limit_c):
    state["last_good"] = record
    return state
```

```
test_first_reading_is_remembered_and_adds_nothing ... ok - OK
```

YELLOW PHASE (REFACTOR)

Verified structure; total correctly remains un-incremented for initial point.